

Compressible Flow (Gas Dynamics)

Equations, Definitions, and Concepts

COMPRESSIBLE FLOW DEFINED

Flow where fluid motion causes changes in the fluid's local thermodynamic properties. By definition, a flow is compressible when $D\rho/Dt \neq 0$, where $D\rho/Dt$ is the substantial derivative of density with respect to time.

ENGINEERING PHYSICS: FUNDAMENTALS

Ideal gases

$$P = \rho RT = mRT/V = n\bar{R}_u T/V$$

Ideal gas constant

$$R = R_u/M_m \quad | \quad m = nM_m$$

Specific heat $c_p - c_v = R \quad | \quad \gamma = c_p/c_v$

$$c_p = \gamma R/(\gamma - 1) \quad | \quad c_v = R/(\gamma - 1)$$

Speed of sound $a = \sqrt{\partial p / \partial \rho}$

Speed of sound (ideal gas)

$$a = \sqrt{\gamma RT} = \sqrt{\gamma P / \rho}$$

Isentropic Processes

Ideal, calorically perfect gases

$$p_2/p_1 = (\rho_2/\rho_1)^\gamma$$

$$p_2/p_1 = (T_2/T_1)^{\gamma/(\gamma-1)}$$

Mass Conservation (steady flow)

$$\sum_i \dot{m}_i = \sum_e \dot{m}_e$$

where $\dot{m} = \rho AV$ (uniform flow)

Substantial Derivative (operator)

$$D/Dt = \partial/\partial t + (\vec{U} \cdot \vec{\nabla})$$

DIMENSIONLESS RATIOS

Knudsen number $Kn = \lambda/L_c$

Kn is the ratio of the molecular mean free path to the flow characteristic length

Mach number $M = U/a$

M is the ratio of the inertial force to the compressibility force

Prandtl number $Pr = \nu/\alpha = \mu c_p/k$

Pr is the ratio of viscous diffusion to thermal diffusion

Reynolds number $Re = \rho UL_c/\mu = UL_c/\nu$

Re is the ratio of the inertial force to the viscous force

Stanton number $St = (Q/A)/[\rho U c_p (T_s - T_g)]$

St is the ratio of the heat transfer to the fluid to the fluid thermal capacity

FLOW REGIMES

Mach number

subsonic, incompressible	$0 < M \leq 0.3$
subsonic, compressible	$0.3 \leq M < 1$
transonic, compressible	$M \approx 1$
sonic, compressible	$M = 1$
supersonic, compressible	$1 < M \leq 5.0$
hypersonic compressible	$M \geq 5.0$

Knudsen number

continuum	$Kn \leq 0.01$
slip flow	$0.01 \leq Kn \leq 0.1$
transition	$0.1 \leq Kn \leq 10$
free molecular	$Kn \geq 10$

Reynold number

internal flow	
laminar	$Re \leq 2,300$
turbulent	$Re \geq 2,300$
external	
laminar	$Re \leq 500,000$
turbulent	$Re \geq 500,000$

ONE-DIMENSIONAL ISENTROPIC FLOW

Ideal gas, calorically perfect; for isentropic flows, stagnation properties are constant

$$\frac{T_o}{T} = 1 + \left(\frac{\gamma - 1}{2}\right) M^2$$

$$\frac{p_o}{p} = \left[1 + \left(\frac{\gamma - 1}{2}\right) M^2\right]^{\gamma/(\gamma-1)}$$

$$\frac{\rho_o}{\rho} = \left[1 + \left(\frac{\gamma - 1}{2}\right) M^2\right]^{1/(\gamma-1)}$$

Critical Conditions ($M^* = 1$)

$$T_o/T^* = (\gamma + 1)/2$$

$$p_o/p^* = [(\gamma + 1) / 2]^{\gamma/(\gamma-1)}$$

$$\rho_o/\rho^* = [(\gamma + 1) / 2]^{1/(\gamma-1)}$$

$$A/A^* = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right) \right]^{(\gamma+1)/2(\gamma-1)}$$

NORMAL SHOCKS

Supersonic flow: perpendicular to freestream

flow direction, non-isentropic, adiabatic

upstream $M_U > 1$, downstream $M_D < 1$

Ideal gas, calorically perfect

$$T_{oU} = T_{oD} \quad | \quad p_{oU} > p_{oD}$$

$$M_D^2 = \frac{(\gamma-1)M_U^2+2}{2\gamma M_U^2-(\gamma-1)}$$

$$\frac{T_D}{T_U} = \frac{a_D^2}{a_U^2} = [2 + (\gamma - 1)M_U^2] \left[\frac{2\gamma M_U^2 - (\gamma - 1)}{(\gamma + 1)^2 M_U^2} \right]$$

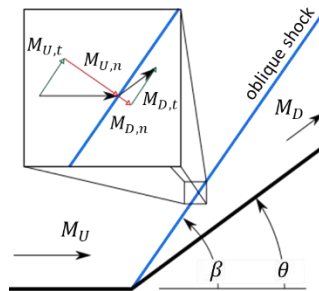
$$\frac{p_D}{p_U} = \left(\frac{2\gamma}{\gamma+1} \right) M_U^2 - \left(\frac{\gamma-1}{\gamma+1} \right)$$

$$\frac{\rho_D}{\rho_U} = (\gamma + 1) M_U^2 / ((\gamma - 1) M_U^2 + 2)$$

$$\frac{p_{oD}}{p_{oU}} = \rho_{oD}/\rho_{oU} = A_U^*/A_D^*$$

$$\frac{p_{oD}}{p_{oU}} = \left[\frac{(\gamma+1)M_U^2}{1 + \frac{\gamma-1}{2}M_U^2} \right]^{\frac{\gamma}{\gamma-1}} \left[\left(\frac{2\gamma}{\gamma+1} M_U^2 - \frac{\gamma-1}{\gamma+1} \right) \right]^{\frac{1}{1-\gamma}}$$

$$s_D - s_U = c_p \ln \left(\frac{T_D}{T_U} \right) - R \ln \left(\frac{p_D}{p_U} \right)$$



An abrupt change in direction produces an oblique shock (supersonic flow).

OBLIQUE SHOCKS

Supersonic flow: shock at an angle to flow direction,

Normal velocity component, normal shock relations hold

Upstream $M_U > 1$, downstream $M_{D,n} < 1$

Due to the tangential velocity component, it is possible to have supersonic flow downstream of an oblique shock

Ideal gas, calorically perfect

$$\tan \theta = \frac{2 \cot \beta [(M_U^2 \sin^2 \beta) - 1]}{\{M_U^2 [\gamma + \cos(2\beta)]\} + 2} \quad \text{Eqn. (OA)}$$

β shock angle (relative to upstream flow)

θ flow turn angle (relative to upstream flow)

$U_{U,t} = U_{D,t}$ tangential velocity components

$U_{U,n}$ is related to $U_{D,n}$ (normal velocity components) via normal shock relations

$$M_{U,n} = M_U \sin \beta \quad M_{U,t} = M_U \cos \beta$$

$$M_{D,n} = M_D \sin(\beta - \theta) \quad M_{D,t} = M_D \cos(\beta - \theta)$$

Solution algorithm

Known (upstream): properties, shock angle

Calculate:

1) θ using Eqn. (OA)

2) Use θ to calc normal and tangential components

3) Use $M_{U,n}$ to calc $M_{D,n}$

4) Use $M_{D,n}$ calc M_D

5) Use relations, such as $T_2 = (T_2/T_1)T_1$, to calc properties downstream

Note: If the turn angle is given instead of the shock angle, must solve Eqn. (OA) for β , then the soln procedure is similar – will end up with two cases (strong shock & weak shock)

PRANDTL-MEYER EXPANSION WAVES

Supersonic flow: flow over a smoothly turning surface – no shocks, series of Mach waves, isentropic process

For a convex surface:

$\uparrow$ speed, $\downarrow$ density, $\downarrow$ pressure, $\downarrow$ temperature

Ideal gas, calorically perfect

$$\nu = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \left[\sqrt{\frac{\gamma+1}{\gamma-1} (M^2 - 1)} \right] - \tan^{-1} [\sqrt{M^2 - 1}]$$

Solution algorithm:

Known (upstream): properties, turn angle

1) $\nu_U(M_U)$

2) $\nu_D(M_D) = \nu_U(M_U) + \theta_D$

3) M_D from $\nu_D(M_D)$; iterate, table, or graph

4) Use isentropic relations for downstream properties

NOZZLES

$\uparrow$ kinetic energy, $\downarrow$ pressure

Enthalpy $h = u + pv$

Mass conservation $\dot{m}_i = \dot{m}_e = \dot{m}$

Cross-sectional area variation

$$dA/A = (M^2 - 1) dU/U$$

subsonic: nozzle area decreases

supersonic: nozzle area increases

sonic: nozzle area minimum (throat)

$$\text{Uniform flow } h_i + 0.5U_i^2 = h_e + 0.5U_e^2$$

Stagnation enthalpy $h_o = h + 0.5U^2$

Uniform, calorically perfect

$$c_p T_i + 0.5U_i^2 = c_p T_e + 0.5U_e^2$$

Back pressure, p_∞ (ambient pressure)

Converging nozzle

For fixed A_e (geometry)

As $p_\infty \downarrow$, $U_e \uparrow$ (subsonic) until $p_\infty = p_e = p^*$

At $p_\infty = p^*$, max U_e , $U_e = a$ ($M_e = 1$)

For $p_\infty < p^*$, $U_e = a$ ($M_e = 1$)

For $p_\infty \leq p^*$ flow is choked

Converging-diverging (de Laval) nozzle

For fixed A_e and A_t (geometry)

Venturi: converging – nozzle; diverging – diffuser

As $p_\infty \downarrow$, $U_e \uparrow$ (subsonic) until $p_t = p^*$

At $p_t = p^*$, max U_t , $U_t = a$ ($M_t = 1$)

For $p_\infty < p^*$, $U_t = a$ ($M_e = 1$)

For $p_\infty \leq p^*$ nozzle throat is choked

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Shock in diverging section

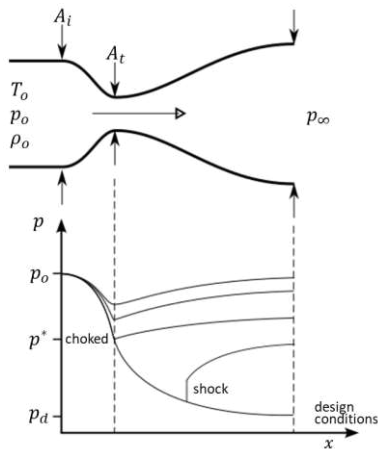
Once $p_t = p^*$, further $\downarrow p_{\infty}$ leads to a shock developing inside the diverging section, and the shock location shifts toward the nozzle exit as p_{∞} continues to $\downarrow$. Presence of shock = non-isentropic. Upstream of shock = nozzle, downstream of shock = diffuser

Design conditions

$p_{\infty} \downarrow$ to the point where there is no shock inside of the nozzle ($p_{\infty} = p_d$), the entire converging-diverging region acts as a nozzle, $M_e > 1$

Off-design conditions

For $p_{\infty} < p_d$, complex flow fields (shocks and expansions) outside of the nozzle: $p_e < p_{\infty}$ (overexpanded), $p_e > p_{\infty}$ (underexpanded)



Pressure variation in a converging-diverging nozzle

MASS FLOW RATE

Ideal gas, calorically perfect

Compressible flow

$$\dot{m} = (A M p_o \sqrt{\frac{\gamma}{R T_o}}) / \left[1 + \frac{(\gamma-1)}{2} M^2 \right]^{(\gamma+1)/2(\gamma-1)}$$

Choked flow

$$\dot{m} = (A p_o \sqrt{\frac{\gamma}{R T_o}}) / \left[\frac{(\gamma+1)}{2} \right]^{(\gamma+1)/2(\gamma-1)}$$

FANNO FLOW

Adiabatic, viscous friction (head loss) – ‘long’ ducts, non-isentropic, steady, one-dimensional, constant cross-sectional area, fully developed, constant friction factor

Friction causes both supersonic and subsonic flows to move toward sonic flow (entropy increase)

$$D_h = 4A/P_{\text{perim}}$$

$$\dot{m}/A = \rho_u U_u = \rho_D U_D \text{ constant mass flux rate}$$

Ideal gas, calorically perfect

$$T_{o,u} = T_{o,D} = T_o^*$$

$$\frac{T_D}{T_U} = (2 + (\gamma-1)M_U^2) / (2 + (\gamma-1)M_D^2)$$

$$\frac{p_D}{p_U} = \frac{M_U}{M_D} \sqrt{\frac{2 + (\gamma-1)M_U^2}{2 + (\gamma-1)M_D^2}}$$

$$\frac{p_{o,D}}{p_{o,U}} = \frac{p_D}{p_U} \left[\frac{2 + (\gamma-1)M_U^2}{2 + (\gamma-1)M_D^2} \right]^{1/(\gamma-1)}$$

$$\frac{T}{T^*} = (\gamma+1)/(2 + (\gamma-1)M^2)$$

$$\frac{p}{p^*} = \frac{1}{M} \sqrt{\frac{\gamma+1}{2 + (\gamma-1)M^2}}$$

$$\frac{\rho}{\rho^*} = \frac{1}{M} \sqrt{\frac{2 + (\gamma-1)M^2}{\gamma+1}}$$

$$\frac{p_o}{p_o^*} = \frac{1}{M} \left[\frac{2 + (\gamma-1)M^2}{\gamma+1} \right]^{(\gamma+1)/2(\gamma-1)}$$

$$f = 8 \left[\left(\frac{8}{Re} \right)^{12} + (C_1 + C_2)^{-1.5} \right]^{1/12} \text{ Churchill eqn.}$$

$$C_1 = \left\{ -2.457 \ln \left[\left(\frac{0.9}{Re} \right) + 0.27 \frac{\epsilon}{D} \right] \right\}^{16}$$

$$C_2 = \left(\frac{37530}{Re} \right)^{16}$$

$$\frac{f L^*}{D_h} = \left(\frac{1-M^2}{\gamma M^2} \right) + \left(\frac{\gamma+1}{2\gamma} \right) \ln \left[\frac{(\gamma+1)M^2}{2 + (\gamma-1)M^2} \right] \text{ Eqn. (FA)}$$

$$\frac{f L}{D_h} = \frac{f L_U}{D_h} - \frac{f L_D}{D_h} \text{ Eqn. (FB)}$$

$$s_D - s_U = c_p \ln \left[\left(\frac{M_D}{M_U} \right)^{(\gamma-1)/\gamma} \left(\frac{1 + \frac{(\gamma-1)}{2} M_U^2}{1 + \frac{(\gamma-1)}{2} M_D^2} \right)^{(\gamma+1)/2\gamma} \right]$$

Solution algorithm

Known (upstream): geometry, properties

Calculate:

- 1) friction factor, f
- 2) $f L_U^*/D_h$ using Eqn. (FA)
- 3) $f L_D^*/D_h$ using Eqn. (FB)
- 4) M_D using Eqn. (FA); iterate, table, or graph
- 5) $T_D = (T_D/T_o)(T_o/T_U)T_U$
- 6) Other properties using Fanno, mass flow rate, or other available relations

RAYLEIGH FLOW

Non-adiabatic (heat transfer to fluid), frictionless (no head loss) – ‘short’ ducts, non-isentropic, steady, one-dimensional (uniform, no boundary layer) flow, constant cross-sectional area, fully developed, constant heat transfer

Heat transfer into the fluid causes both supersonic and subsonic flows to move toward sonic flow (entropy increase)

$$\dot{m}/A = \rho_u U_u = \rho_D U_D \text{ constant mass flux rate}$$

Ideal gas, calorically perfect

Max heat transfer (to sonic conditions)

$$q_{\text{max}} = \dot{Q}_{\text{max}}/\dot{m} = c_p(T_o^* - T_{o,u})$$

Heat transfer

$$q = \dot{Q}/\dot{m} = c_p(T_{o,D} - T_{o,u}) \text{ Eqn. (RA)}$$

Property relations

$$\frac{T_D}{T_U} = \left(\frac{M_D}{M_U} \right) \left[\frac{(1 + \gamma M_U^2)/(1 + \gamma M_D^2)}{1} \right]$$

$$\frac{p_D}{p_U} = (1 + \gamma M_U^2)/(1 + \gamma M_D^2)$$

$$\frac{T}{T^*} = \left[\frac{(\gamma+1)M}{(1 + \gamma M^2)} \right]^2$$

$$\frac{p}{p^*} = (\gamma+1)/(1 + \gamma M^2)$$

$$\frac{\rho}{\rho^*} = \frac{U^*}{U} = (1 + \gamma M^2)/(1 + \gamma M^2)$$

$$\frac{T_o}{T_o^*} = \frac{2(1+\gamma)M^2}{(1+\gamma M^2)^2} \left(1 + \frac{\gamma-1}{2} M^2 \right) \text{ Eqn. (RB)}$$

$$\frac{p_o}{p_o^*} = \frac{1+\gamma}{1+\gamma M^2} \left[\frac{2 + (\gamma-1)M^2}{1+\gamma} \right]^{1/(\gamma-1)}$$

$$s_D - s_U = c_p \ln \left[\left(\frac{M_D}{M_U} \right)^2 \left(\frac{1 + \gamma M_U^2}{1 + \gamma M_D^2} \right)^{(\gamma+1)/\gamma} \right]$$

Solution algorithm

Known (upstream): properties, heat transfer

Calculate:

- 1) Any unknown upstream properties required (may use isentropic relations here)
- 2) $T_{o,D}$ using Eqn. (RA)
- 3) $T_{o,u}/T_o^*$ using Eqn. (RB)

$$4) T_{o,D}/T_o^* = (T_{o,D}/T_{o,u})(T_{o,u}/T_o^*)$$

5) M_D using Eqn. (RB); iterate, table, or graph

6) Other downstream properties using Rayleigh flow, mass flow, and other relations.

HYPERSONIC FLOW

Calorically imperfect

Dissociation/chemical reactions

Rarefied: velocity slip/temperature jump

Thermal radiation heat transfer

Thin shock layer

Viscous dissipation/heating

DEFINITIONS

Adiabatic: no heat transfer

Calorically perfect: constant specific heats

Choked flow: A decrease in downstream pressure will not affect the mass flow rate through a restriction at sonic flow conditions because sound cannot propagate upstream under these conditions.

Critical (sonic) conditions: internal flows where $Ma = 1$

Eulerian frame of reference: observing the changes in the flow over time at a given point (or region) in space

Isentropic: constant entropy process

Lagrangian frame of reference: observing the changes in an infinitesimal control mass (fluid parcel) over time as it moves along with the flow through space

Shock: a discontinuity in supersonic flow properties resulting from pressure wave interactions and accompanied by an abrupt decrease in flow speed (and increase in entropy)

Stagnation (total) properties: thermodynamic properties that would exist if the flow were decelerated to zero velocity following a frictionless & adiabatic (thus isentropic) process.

Substantial derivative (also material or total derivative): rate of change of thermodynamic properties of a control mass (fluid parcel) as it moves along with the flow – a derivative in a Lagrangian reference frame

Steady: variables do not change over time

Thermally perfect: internal energy, enthalpy, and specific heat are a function of temperature only; ideal gases are thermally perfect

Uniform flow: constant cross-sectional variables (no viscous effects)

USEFUL CONSTANTS

Ideal gas constant (air) $R_{\text{air}} = 287 \text{ J/kg K}$

Universal gas constant $R_u = 8.314 \text{ J/mol K}$

Specific heat ratio (calorically perfect; diatomic gas, also valid for air) $\gamma_{\text{dia}} = \gamma_{\text{air}} = 1.4$

Specific heat ratio (calorically perfect; monoatomic gases) $\gamma_{\text{mono}} = 1.4$

USEFUL RESOURCE

Ames Research Staff, “Equations, Tables, and Charts for Compressible Flow,” NACA-TR-1135, 1953. [https://ntrs.nasa.gov/citations/19930091059]